

Quant II

Lab 10

Yinxuan Wang

2026-04-24

Today's Agenda

- Mediation
- Moderation
- Interference

Mediation: concepts review

- Total Effect: $\tau_i = Y_i(1, M_i(1)) - Y_i(0, M_i(0))$
 - Overall effect of treatment on outcome

Mediation: concepts review

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 - Overall effect of treatment on outcome
- Natural Direct Effect: $\zeta_i(t) = Y_i(1, M_i(t)) - Y_i(0, M_i(t))$
 - Share of total effect that doesn't go through M.

Mediation: concepts review

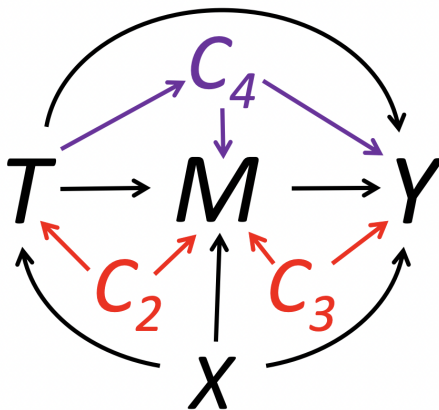
- Total Effect: $\tau_i = Y_i(1, M_i(1)) - Y_i(0, M_i(0))$
 - Overall effect of treatment on outcome
- Natural Direct Effect: $\zeta_i(t) = Y_i(1, M_i(t)) - Y_i(0, M_i(t))$
 - Share of total effect that doesn't go through M.
- Natural Mediation Effect: $\delta_i(t) = Y_i(t, M_i(1)) - Y_i(t, M_i(0))$
 - Share of total effect that goes through M.
- $\tau_i = \delta_i(t) + \zeta_i(1 - t)$
- ID assumption for δ and ζ : sequential ignorability

Sequential ignorability

$$\begin{aligned}T_i &\perp (Y_i(t', m), M_i(t)) | X_i = x \\M_i(t) &\perp Y_i(t', m) | T_i, X_i = x\end{aligned}$$

- First part: CIA, satisfied in a randomized experiment
- Second part: no omitted post-treatment confounder or other mediator causally connected to M

Sequential ignorability



Causal mediation analysis

- R package `mediation`: causal mediation analysis under sequential ignorability
- Identify effect of T on M given X and the effect of M on Y given T and X
- With them compute the direct/mediation effects
- In the special case of linear models one can multiply the coefficients

Causal mediation analysis

- Working example from the `mediation` package: Brader et al (2008)
- T : Media stories about immigration
- Y : Letter about immigration policy to representative in Congress
- M : Anxiety
- X : Age, education, gender, income

Causal mediation analysis

```
library(mediation)

data(framing)

set.seed(2014)

# Model for the mediator (T + X)
med.fit <- lm(emo ~ treat + age + educ + gender + income, data = framing)

# Model for the outcome (M + T + X)
out.fit <- glm(cong_mesg ~ emo + treat + age + educ + gender + income, data = framing,
               family = binomial("probit"))

# Compute the mediation effects
med.out <- mediate(med.fit, out.fit, treat = "treat", mediator = "emo",
                  robustSE = TRUE, sims = 1000)
```

Causal mediation analysis

```
summary(med.out)
```

Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	0.082588	0.035643	0.143986	<2e-16 ***
ACME (treated)	0.083121	0.034797	0.144589	<2e-16 ***
ADE (control)	0.013695	-0.096655	0.131807	0.818
ADE (treated)	0.014228	-0.110099	0.136483	0.818
Total Effect	0.096816	-0.029035	0.230085	0.136
Prop. Mediated (control)	0.770623	-6.396763	4.704882	0.136
Prop. Mediated (treated)	0.793825	-5.750632	4.515358	0.136
ACME (average)	0.082855	0.035104	0.144353	<2e-16 ***
ADE (average)	0.013961	-0.104708	0.134339	0.818
Prop. Mediated (average)	0.782224	-6.073697	4.610120	0.136

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 265

Treatment Mediator Interaction

```
med.interact.fit <- lm(emo ~ treat + age + educ + gender + income, data=framing)
out.interact.fit <- glm(cong_mesg ~ emo * treat + age + educ + gender + income,
  data = framing, family = binomial("probit"))
med.interact.out <- mediate(med.interact.fit, out.interact.fit,
  treat = "treat", mediator = "emo",
  robustSE = TRUE, sims = 1000)
summary(med.interact.out)
```

Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	0.0749872	0.0274878	0.1298461	<2e-16 ***
ACME (treated)	0.0967012	0.0363125	0.1699029	<2e-16 ***
ADE (control)	-0.0038392	-0.1141123	0.1179175	0.936
ADE (treated)	0.0178749	-0.1113305	0.1503177	0.792
Total Effect	0.0928620	-0.0449654	0.2294911	0.206
Prop. Mediated (control)	0.6739715	-6.9158511	5.8224364	0.206
Prop. Mediated (treated)	0.9153167	-7.5107069	7.4853944	0.206
ACME (average)	0.0858442	0.0350753	0.1413974	<2e-16 ***
ADE (average)	0.0070178	-0.1087397	0.1320763	0.910
Prop. Mediated (average)	0.7946441	-6.9514609	6.7444955	0.206

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 265

Simulations: 1000

Treatment Mediator Interaction

```
test.TMint(med.interact.out, conf.level = .95)
```

Test of $ACME(1) - ACME(0) = 0$

data: estimates from med.interact.out

$ACME(1) - ACME(0) = 0.021714$, p-value = 0.386

alternative hypothesis: true $ACME(1) - ACME(0)$ is not equal to 0

95 percent confidence interval:

-0.02801036 0.07915849

Sensitivity Analysis for SI

```
sens.out <- medsens(med.out, rho.by = 0.1, effect.type = "indirect", sims = 1000)
summary(sens.out)
```

Mediation Sensitivity Analysis: Average Mediation Effect

Sensitivity Region: ACME for Control Group

	Rho	ACME(control)	95% CI Lower	95% CI Upper	$R^2_M \cdot R^2_{Y*}$	$R^2_M - R^2_{Y-}$
[1,]	0.3	0.0055	-0.0076	0.0172	0.09	0.0493
[2,]	0.4	-0.0094	-0.0265	0.0020	0.16	0.0877

Rho at which ACME for Control Group = 0: 0.3

$R^2_M \cdot R^2_{Y*}$ at which ACME for Control Group = 0: 0.09

$R^2_M - R^2_{Y-}$ at which ACME for Control Group = 0: 0.0493

Sensitivity Region: ACME for Treatment Group

	Rho	ACME(treated)	95% CI Lower	95% CI Upper	$R^2_M \cdot R^2_{Y*}$	$R^2_M - R^2_{Y-}$
[1,]	0.3	0.0064	-0.0086	0.020	0.09	0.0493
[2,]	0.4	-0.0113	-0.0319	0.002	0.16	0.0877

Rho at which ACME for Treatment Group = 0: 0.3

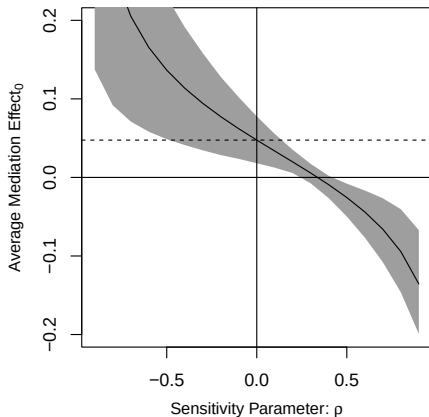
$R^2_M \cdot R^2_{Y*}$ at which ACME for Treatment Group = 0: 0.09

$R^2_M - R^2_{Y-}$ at which ACME for Treatment Group = 0: 0.0493

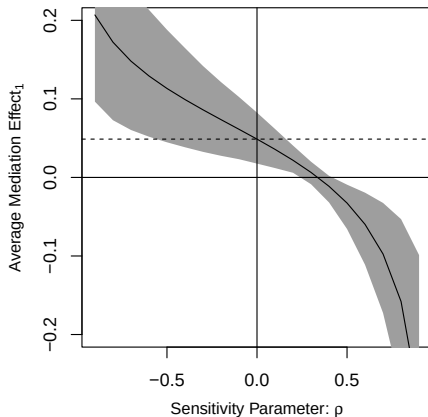
Sensitivity Analysis for SI

```
par(mfrow = c(1, 2))  
plot(sens.out, sens.par = "rho", main = "Anxiety", ylim = c(-0.2, 0.2))
```

Anxiety



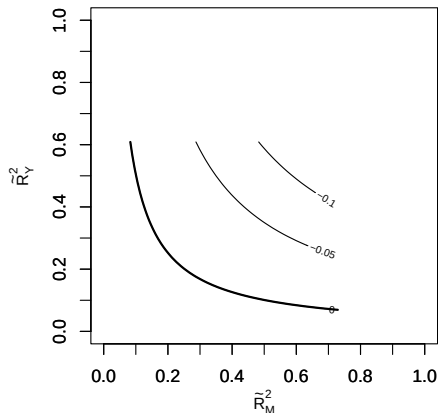
Anxiety



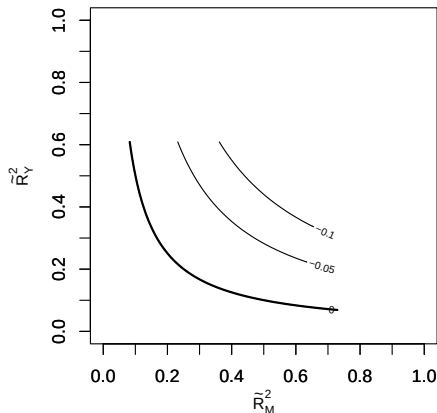
Sensitivity Analysis for SI

```
par(mfrow = c(1, 2))  
plot(sens.out, sens.par = "R2", r.type = "total", sign.prod = "positive")
```

$ACME_0(\tilde{R}_M^2, \tilde{R}_Y^2), \text{sgn}(\lambda_2\lambda_3) = 1$



$ACME_1(\tilde{R}_M^2, \tilde{R}_Y^2), \text{sgn}(\lambda_2\lambda_3) = 1$



Controlled Direct Effect

- Controlled Direct Effect: $\kappa_i(m) = Y_i(1, m) - Y_i(0, m)$
- Effect of T on Y when M has the same value for all units.
- Relative to NDE and NME, can be identified under assumptions that allow observed intermediate confounders.

Controlled Direct Effect

- Controlled Direct Effect: $\kappa_i(m) = Y_i(1, m) - Y_i(0, m)$
- Effect of T on Y when M has the same value for all units.
- Relative to NDE and NME, can be identified under assumptions that allow observed intermediate confounders.
- A natural approach to close M channel:
 - Include M as control in the regression
 - In presence of intermediate confounders, this introduces post-treatment bias
- CDE is an estimand that allows to overcome this issue

Sequential g-estimation

- Popularized in political science by Acharya, Blackwell, and Sen (2016)
- Procedure in two stages:
 - Regress Y on M , T , pre-treatment and intermediate variables
 - Subtract from Y the effect of M : get the “demediated” Y ,
$$\tilde{Y} = Y - \hat{\beta}_M M$$
 - Regress $\tilde{Y}$ on T and pre-treatment variables
- Doing it by hand would ignore variability in $\tilde{Y}$, which is an estimated quantity, resulting in wrong SEs
- Use the package `DirectEffects` or bootstrap
- Center the mediator at the value you want to “fix” it at

Sequential g-estimation

- Why is it important?
- Research questions may involve comparisons that “hold fixed” things realized after the treatment

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- Do natural shocks impact political development even in absence of physical destruction?
- Does ethnic diversity lead to conflict even in absence of government instability?

Sequential g-estimation

- Why is it important?
- Research questions may involve comparisons that “hold fixed” things realized after the treatment
- Do natural shocks impact political development even in absence of physical destruction?
- Does ethnic diversity lead to conflict even in absence of government instability?
- We may also want to rule causal mechanisms alternative to our theory
 - Are the effects of slavery/famine just due to subsequent changes in racial/ethnic composition? (Acharya, Blackwell, and Sen (2016); Rozenas and Zhukov (2019) resp.)

ACDE Example

- Alesina, Giuliano, and Nunn (2013): data provided with the DirectEffects package
- Y : share of political positions held by women in 2000
- T_i : relative proportion of ethnic groups that traditionally used the plow within a country
- M_i : log GDP per capita in 2000, mean-centered
- Z_i : post-treatment, pre-mediator intermediate confounders
 - civil conflict, interstate conflict, oil, European descent, communist, polity2..)
- X_i : pre-treatment characteristics of the country
 - tropical climate, agricultural suitability, large animals, political hierarchies, economic complexity, rugged

ACDE Example

```
library(DirectEffects)

data("ploughs")

# adjusted OLS benchmark
mod <- lm(women_politics ~ plow + agricultural_suitability +
          tropical_climate + large_animals +
          political_hierarchies + economic_complexity + rugged,
          data = ploughs)
```


ACDE Example

```
summary(mod)
```

Call:

```
lm(formula = women_politics ~ plow + agricultural_suitability +  
    tropical_climate + large_animals + political_hierarchies +  
    economic_complexity + rugged, data = ploughs)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.060	-5.292	-2.000	3.752	25.923

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	17.5604	4.9038	3.581	0.000466	***
plow	-2.1032	2.1270	-0.989	0.324422	
agricultural_suitability	0.9712	2.5761	0.377	0.706721	
tropical_climate	-7.5137	1.9621	-3.830	0.000191	***
large_animals	-8.7104	4.0510	-2.150	0.033201	*
political_hierarchies	0.9359	0.9722	0.963	0.337344	
economic_complexity	1.0284	0.5430	1.894	0.060202	.
rugged	-0.6370	0.4961	-1.284	0.201137	

Sig. if. < .05. (2-tailed). .001. (2-tailed). .01. (2-tailed). .05. (2-tailed). .1. (2-tailed). 1. (2-tailed).

ACDE Example

```
## Formula for sequential_g
form_main <- women_politics ~ plow + agricultural_suitability +
  tropical_climate + large_animals + political_hierarchies +
  economic_complexity + rugged | # pre-treatment vars
  years_civil_conflict + years_interstate_conflict +
  oil_pc + european_descent + communist_dummy +
  polity2_2000 + serv_va_gdp2000 | # intermediate vars
  centered_ln_inc + centered_ln_incsq # mediating vars

## Sequential g-estimation
direct <- sequential_g(formula = form_main, data = ploughs)
```

ACDE Example

```
summary(direct)
```

t test of coefficients:

	Estimate	Std. Err.	t value	Pr(> t)	
(Intercept)	12.18450	3.64442	3.3433	0.001121	**
plow	-4.83879	2.34467	-2.0637	0.041312	*
agricultural_suitability	4.57388	3.10477	1.4732	0.143458	
tropical_climate	-2.18919	2.10505	-1.0400	0.300554	
large_animals	-1.33001	3.40008	-0.3912	0.696401	
political_hierarchies	0.49575	1.09060	0.4546	0.650283	
economic_complexity	-0.10521	0.42973	-0.2448	0.807029	
rugged	-0.30869	0.47821	-0.6455	0.519888	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ACDE Example

```
# bootstrap standard errors
out.boots <- boots_g(direct, boots = 1000)
summary(out.boots)
```

Summary of bootstrapped coefficients:

	Estimate	Std. Err.	t value	2.5 %	97.5 %
(Intercept)	12.74099	4.36300	2.92024	4.52462	21.7429
plow	-5.18375	2.87190	-1.80499	-11.17429	-0.1333
agricultural_suitability	4.24061	3.28983	1.28901	-2.45052	10.6184
tropical_climate	-2.35600	2.40949	-0.97780	-7.03411	2.4693
large_animals	-1.14330	4.82772	-0.23682	-11.25269	7.6029
political_hierarchies	0.60507	1.15239	0.52506	-1.63297	2.8518
economic_complexity	-0.18219	0.54215	-0.33605	-1.23135	0.8571
rugged	-0.29143	0.51660	-0.56414	-1.27393	0.7947

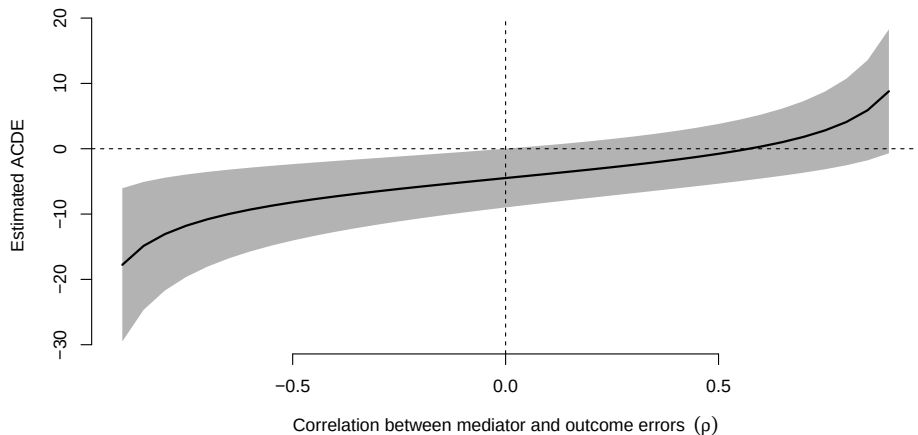
ACDE Sensitivity Analysis

Only supports one mediator

```
direct1 <- sequential_g(women_politics ~ plow +
                        agricultural_suitability + tropical_climate +
                        large_animals + political_hierarchies +
                        economic_complexity + rugged | # pre-treatment vars
                        years_civil_conflict + years_interstate_conflict +
                        oil_pc + european_descent + communist_dummy +
                        polity2_2000 + serv_va_gdp2000 | # intermediate vars
                        centered_ln_inc, # one mediator
                        ploughs)
out_sens <- cdesens(direct1, var = "plow")
```

ACDE Sensitivity Analysis

```
plot(out_sens)
```



- Characterize treatment effect heterogeneity
 - Going beyond the aggregation
 - e.g., for a policy intervention, on what sub-populations the intervention is more effective

Issues with linear interaction terms

- Linear interaction terms used to study how the treatment effect evolves over the distribution of the moderator

Issues with linear interaction terms

- Linear interaction terms used to study how the treatment effect evolves over the distribution of the moderator
- Hainmueller, Mummolo, and Xu (2019) point out that this practice relies on requirements which might be violated
- TE changes linearly in the moderator at any point of its distribution
 - May be non-linear or non-monotonic
- Common support between treatment and moderator
 - If not, the model relies on linear extrapolation

Working example

Slaveholding and state-building in the US South

Slavery, Reconstruction, and Bureaucratic Capacity in the American South

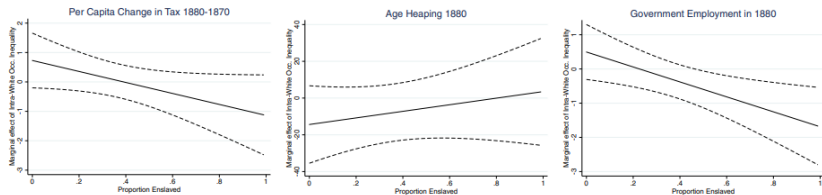
PAVITHRA SURYANARAYAN *Johns Hopkins University*

STEVEN WHITE *Syracuse University*

Conventional political economy models predict taxation will increase after franchise expansion to low-income voters. Yet, contrary to expectations, in ranked societies—where social status is a cleavage—elites can instead build cross-class coalitions to undertake a strategy of bureaucratic weakening to limit future redistributive taxation. We study a case where status hierarchies were particularly extreme: the post-Civil War American South. During Reconstruction, under federal oversight, per capita taxation was higher in counties where slavery had been more extensive before the war, as predicted by standard theoretical models. After Reconstruction ended, however, taxes fell and bureaucratic capacity was weaker where slavery had been widespread. Moreover, higher intrawhite economic inequality was associated with lower taxes and weaker capacity after Reconstruction in formerly high-slavery counties. These findings on the interaction between intrawhite economic inequality and pre-War slavery suggest that elites built cross-class coalitions against taxation where whites sought to protect their racial status.

Moderation with linear estimator

FIGURE 7. Marginal Effect of Intrawhite Inequality on Taxation and Bureaucratic Quality



interflex

- interflex package (in both R and Stata) proposes a more flexible procedure to moderation, proposed by Hainmueller, Mummolo, and Xu (2019)

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Binning estimator

- Divide the support of Z into j bins (e.g. terciles), indicated by G_j , and estimate

$$y_{ij} = \sum_{j=1}^J \{ \alpha_j + \beta_j D_{ij} + \gamma_j (Z_{ij} - Z_j^M) + \delta_j (Z_{ij} - Z_j^M) D_{ij} \} G_j + \psi X_{ij} + \epsilon_{ij}$$

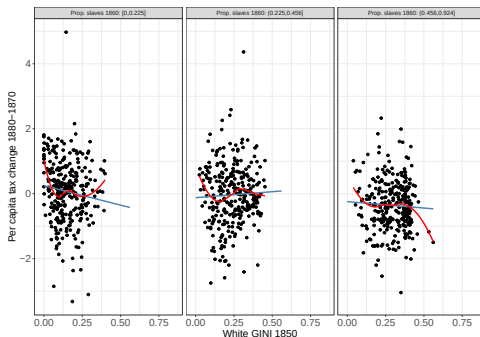
- Z_j^M is the median value of Z inside bin j . Given the specification, β_j s are the conditional ATEs at the center of each bin.

Moderation using binning estimator

```
library(interflex); library(haven); library(tidyverse)

# Import the data
d <- read_dta("suri_white_preprocessed.dta") %>% as.data.frame()

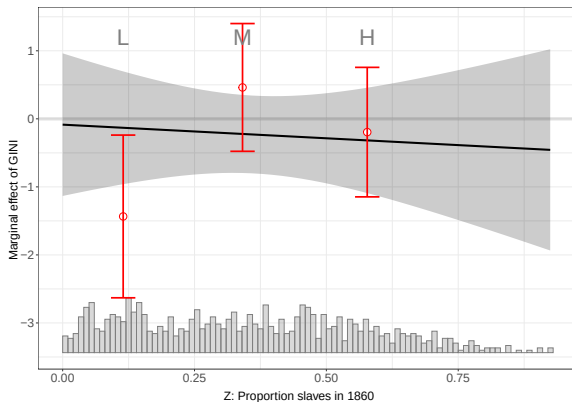
# Raw data plot
# Note we are not including controls used in the paper
interflex(estimator = "raw", data = d, Y = "tax_diff", D = "county_sei_gini_whitemale_1850",
          X = "pslave1860", ylab = "Marginal effect of GINI",
          xlab = "Z: Proportion slaves in 1860", theme.bw = T, ncols=3,
          Dlabel = "White GINI 1850", Ylabel = "Per capita tax change 1880-1870",
          Xlabel = "Prop. slaves 1860")
```



Moderation using the binning estimator

```
# Heterogeneous TE with interflex (note in this notation Z and X are inverted)
out <- interflex(estimator = "binning", data = d,
  Y = "tax_diff", D = "county_sei_gini_whitemale_1850",
  X = "pslave1860", ylab = "Marginal effect of GINI",
  xlab = "Z: Proportion slaves in 1860", theme.bw = T)
```

```
out$figure
```



Kernel estimator

- Allow TE to vary over the whole distribution of the moderator, estimating the following semiparametric model

$$y_i = f(Z_i) + g(Z_i)D_i + h(Z_i)X_i + \epsilon_i$$

Moderation with the kernel estimator

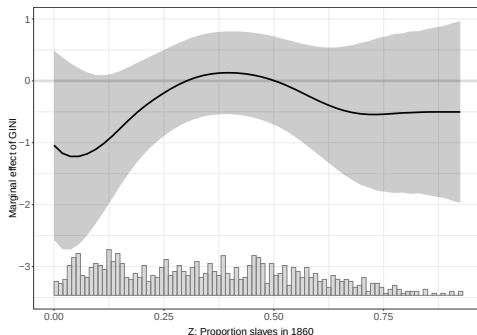
```
set.seed(123)
outk <- interflex(estimator = "kernel", data = d,
  Y = "tax_diff", D = "county_sei_gini_whitemale_1850",
  X = "pslave1860", ylab = "Marginal effect of GINI",
  xlab = "Z: Proportion slaves in 1860", theme.bw = T)
```

Cross-validating bandwidth ...

Optimal bw=0.3042.

Number of evaluation points:50

```
outk$figure
```



Diagnostic tools

- `interflex` also gives diagnostic tools for model specification
- E.g. Wald tests for the hypothesis that the simple linear interaction is correct
- With a slight reparametrization, the null hypothesis is
 - bin-specific deviations from the reference bin are jointly zero
 - i.e. constant coefficients

Diagnostic tools

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```
out$tests$p.wald
```

```
[1] "0.151"
```

Effect Modification and Causal Moderation

Factorial Difference-in-Differences

- Yiqing Xu, Anqi Zhao, and Peng Ding (2026)
- FDID distinguishes effect modification from causal moderation

Effect Modification and Causal Moderation

Factorial Difference-in-Differences

- Yiqing Xu, Anqi Zhao, and Peng Ding (2026)
- FDID distinguishes effect modification from causal moderation
- Standard DID assumptions identify effect modification, not causal moderation
- In the FDID setting, interpreting effect modification as causal moderation requires an additional factorial parallel-trends assumption (could be conditional on observed covariates)
- Slides https://yiqingxu.org/papers/2025_fdid/fdid_slides.pdf
- Paper <https://doi.org/10.1080/01621459.2026.2628343>

Interference

SUTVA violations

- Potential outcomes might depend on others' treatment assignment.
- Example: a voter mobilization message may affect not only the contacted voter, but also the voter's household member or network peers.
- In network experiments, we summarize others' treatment assignments through an **exposure mapping**.

Design-Based Inference with Interference

Zonszein, Samii, and Aronow interference package

- Treatment is randomly assigned.
- Units are connected through a known network.
- Define exposure conditions, such as:
 - directly treated;
 - indirectly exposed through treated neighbors;
 - neither directly nor indirectly exposed.
- Estimation uses known randomization probabilities for each exposure condition.

Design-Based Inference with Interference

Zonszein, Samii, and Aronow interference package

- Treatment is randomly assigned.
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- Define exposure conditions, such as:
 - directly treated;
 - indirectly exposed through treated neighbors;
 - neither directly nor indirectly exposed.
- Estimation uses known randomization probabilities for each exposure condition.

```
# devtools::install_github('szonszein/interference')  
library(interference)
```

Simulation

```
N <- 50
p <- 0.2

# Randomly assign 20% of units to treatment
Z <- make_tr_vec_permutation(N, p, R = 1, seed = 42)

# Generate a small-world network
adj_matrix <- make_adj_matrix(
  N, model = "small_world", seed = 42)
```

Exposure Mapping

Define exposure by own treatment and neighbors' treatment.

- `isol_dir`: directly treated, no treated neighbor.
- `dir_ind1`: directly treated and indirectly exposed through a treated neighbor.
- `ind1`: untreated but has a treated neighbor.
- `no`: untreated and has no treated neighbor.

```
D <- make_exposure_map_AS(  
  adj_matrix, Z, hop = 1)
```

```
D
```

	<code>dir_ind1</code>	<code>isol_dir</code>	<code>ind1</code>	<code>no</code>
[1,]	0	0	0	1
[2,]	0	0	1	0
[3,]	0	0	1	0
[4,]	0	0	1	0
[5,]	0	0	1	0
[6,]	0	0	1	0
[7,]	1	0	0	0
[8,]	0	0	1	0
[9,]	1	0	0	0

Treatment Assignment Probabilities

To estimate causal effects, we need the probability that each unit falls into each exposure condition.

```
omega <- make_tr_vec_permutation(  
  N, p, R = 50, seed = 42,  
  allow_repetitions = FALSE)  
  
prob_exposure <- make_exposure_prob(  
  omega, adj_matrix, make_exposure_map_AS, list(hop = 1))  
  
make_prob_exposure_cond(prob_exposure)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
dir_ind1	0.1176471	0.1568627	0.09803922	0.09803922	0.05882353	0.07843137
isol_dir	0.1372549	0.1372549	0.21568627	0.15686275	0.13725490	0.13725490
ind1	0.2745098	0.3137255	0.45098039	0.58823529	0.31372549	0.56862745
no	0.5294118	0.4509804	0.29411765	0.21568627	0.54901961	0.27450980
	[,7]	[,8]	[,9]	[,10]	[,11]	[,12]
dir_ind1	0.13725490	0.19607843	0.1568627	0.23529412	0.07843137	0.1176471
isol_dir	0.03921569	0.03921569	0.1372549	0.05882353	0.07843137	0.1176471
ind1	0.58823529	0.47058824	0.5686275	0.52941176	0.60784314	0.6470588
no	0.29411765	0.35294118	0.1960784	0.23529412	0.29411765	0.1764706
	[,13]	[,14]	[,15]	[,16]	[,17]	[,18]

Observed Outcomes under Interference

```
potential_outcomes <- make_dilated_out(  
  adj_matrix, make_corr_out, seed = 1101,  
  multipliers = NULL, hop = 1)  
  
observed_outcomes <- rowSums(D * t(potential_outcomes))  
  
observed_outcomes
```

```
[1] 45.970326  9.229195 39.199710 221.697102 27.551556 49.150119  
[7] 27.293816 11.972031 119.792919 72.175516 22.802367 128.673026  
[13] 67.847264 41.477358 81.919395 134.006797 68.534496 45.068685  
[19] 21.040936 37.249676 66.956477 7.578687 45.843180 18.051933  
[25] 6.794359 44.013584 27.225227 49.743773 204.823393 301.411684  
[31] 54.928201 64.781268 17.592564 37.164950 30.377814 21.711319  
[37] 230.315254 90.244504 124.361504 207.631365 69.292506 29.769166  
[43] 14.415319 7.827460 110.696577 65.251477 104.392538 166.648398  
[49] 50.357569 16.363472
```

Estimate Spillover Effects

```
estimates <- estimates(  
  D, observed_outcomes, prob_exposure,  
  control_condition = "no")  
estimates$tau_ht
```

```
dir_ind1  isol_dir      ind1  
-31.21724 193.98273    19.86354
```

```
estimates$var_tau_h
```

```
dir_ind1  isol_dir      ind1  
1448.7279 42576.0153    762.0521
```